



# Probability with a Defined Sample Space

**Grades 9-12 • SAT / ACT mathematics foundations**

Problem-Solving and Data Analysis • Compute probabilities from equally likely outcomes.

Original focused skill practice for learners preparing for college-entrance mathematics. Includes models, 16 practice tasks, six explain-and-check tasks, a four-item exit ticket, and worked solutions. Not a full practice exam.

**PRINT • TEACH • REVISIT**



# Plan a focused study session

## Learning objective

Compute probabilities from equally likely outcomes.

## Readiness

Use after learners have studied the relevant algebra, geometry, or data concepts. Grade tags describe a typical preparation audience; readiness depends on prior coursework.

## Suggested use

Session 1: 10 minutes on worked examples and 20–30 minutes on practice. Session 2: review errors, then use explain-and-check tasks. Finish with the independent exit ticket. Pause and reteach a prerequisite when the model is not yet understood.

## What this resource covers

The named mathematics skill is useful within SAT and ACT preparation. The packet does not cover every exam domain, reproduce an official section, provide standardized timing, or predict a score.

## Materials

Pencil, paper, and a calculator for checking when useful. Practice both symbolic reasoning and calculator verification. Follow each exam publisher's current policies for an actual test.

## Student directions

Read the entire question, define quantities, show a method, and check the result. For a written explanation, cite the calculation or feature that supports your conclusion.



## Two worked examples

**Model 1. A bag contains 4 red and 8 blue identical-size counters. Two are drawn without replacement. Find the probability of red first and blue second.**

$P(\text{red first}) = 4/12$ . Then 8 blue remain among 11 counters. Multiply:  $4/12 \times 8/11 = 8/33$ .

**Model 2. A bag contains 5 red and 4 blue identical-size counters. Two are drawn without replacement. Find the probability of red first and blue second.**

$P(\text{red first}) = 5/9$ . Then 4 blue remain among 8 counters. Multiply:  $5/9 \times 4/8 = 5/18$ .

Pause before looking at a solution: which relationship, definition, or constraint determines the first step?

## Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. A bag contains 8 red and 7 blue identical-size counters. Two are drawn without replacement. Find the probability of red first and blue second.

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2. A bag contains 2 red and 4 blue identical-size counters. Two are drawn without replacement. Find the probability of red first and blue second.

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3. A bag contains 2 red and 7 blue identical-size counters. Two are drawn without replacement. Find the probability of red first and blue second.

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## Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

4. A bag contains 7 red and 8 blue identical-size counters. Two are drawn without replacement. Find the probability of red first and blue second.

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5. A bag contains 8 red and 8 blue identical-size counters. Two are drawn without replacement. Find the probability of red first and blue second.

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6. A bag contains 7 red and 3 blue identical-size counters. Two are drawn without replacement. Find the probability of red first and blue second.

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## Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

7. A bag contains 7 red and 4 blue identical-size counters. Two are drawn without replacement. Find the probability of red first and blue second.

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8. A bag contains 9 red and 5 blue identical-size counters. Two are drawn without replacement. Find the probability of red first and blue second.

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9. A bag contains 3 red and 2 blue identical-size counters. Two are drawn without replacement. Find the probability of red first and blue second.

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## Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

10. A bag contains 3 red and 6 blue identical-size counters. Two are drawn without replacement. Find the probability of red first and blue second.

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11. A bag contains 8 red and 8 blue identical-size counters. Two are drawn without replacement. Find the probability of red first and blue second.

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12. A bag contains 5 red and 7 blue identical-size counters. Two are drawn without replacement. Find the probability of red first and blue second.

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## Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

13. A bag contains 4 red and 8 blue identical-size counters. Two are drawn without replacement. Find the probability of red first and blue second.

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14. A bag contains 6 red and 4 blue identical-size counters. Two are drawn without replacement. Find the probability of red first and blue second.

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15. A bag contains 2 red and 4 blue identical-size counters. Two are drawn without replacement. Find the probability of red first and blue second.

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## Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

16. A bag contains 5 red and 7 blue identical-size counters. Two are drawn without replacement. Find the probability of red first and blue second.

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# Explain and check • Transfer practice

Solve the task, then explain a check or limitation. Compare your reasoning with the worked key only after making your own attempt.

1. A bag contains 7 red and 3 blue identical-size counters. Two are drawn without replacement. Find the probability of red first and blue second.

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2. A bag contains 7 red and 5 blue identical-size counters. Two are drawn without replacement. Find the probability of red first and blue second.

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# Explain and check • Transfer practice

Solve the task, then explain a check or limitation. Compare your reasoning with the worked key only after making your own attempt.

3. A bag contains 5 red and 5 blue identical-size counters. Two are drawn without replacement. Find the probability of red first and blue second.

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4. A bag contains 2 red and 3 blue identical-size counters. Two are drawn without replacement. Find the probability of red first and blue second.

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# Explain and check • Transfer practice

Solve the task, then explain a check or limitation. Compare your reasoning with the worked key only after making your own attempt.

5. A bag contains 3 red and 3 blue identical-size counters. Two are drawn without replacement. Find the probability of red first and blue second.

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6. A bag contains 9 red and 6 blue identical-size counters. Two are drawn without replacement. Find the probability of red first and blue second.

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# Exit ticket • Independent check

Complete independently. Show reasoning and any relevant restriction. This is a classroom learning check, not a standardized test score.

1. A bag contains 7 red and 6 blue identical-size counters. Two are drawn without replacement. Find the probability of red first and blue second.

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2. A bag contains 6 red and 2 blue identical-size counters. Two are drawn without replacement. Find the probability of red first and blue second.

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## Exit ticket • Independent check

Complete independently. Show reasoning and any relevant restriction. This is a classroom learning check, not a standardized test score.

3. A bag contains 9 red and 3 blue identical-size counters. Two are drawn without replacement. Find the probability of red first and blue second.

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4. A bag contains 7 red and 5 blue identical-size counters. Two are drawn without replacement. Find the probability of red first and blue second.

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# Practice worked solutions

## 1. $\frac{4}{15}$

P(red first) =  $\frac{8}{15}$ . Then 7 blue remain among 14 counters. Multiply:  $\frac{8}{15} \times \frac{7}{14} = \frac{4}{15}$ .

## 2. $\frac{4}{15}$

P(red first) =  $\frac{2}{6}$ . Then 4 blue remain among 5 counters. Multiply:  $\frac{2}{6} \times \frac{4}{5} = \frac{4}{15}$ .

## 3. $\frac{7}{36}$

P(red first) =  $\frac{2}{9}$ . Then 7 blue remain among 8 counters. Multiply:  $\frac{2}{9} \times \frac{7}{8} = \frac{7}{36}$ .

## 4. $\frac{4}{15}$

P(red first) =  $\frac{7}{15}$ . Then 8 blue remain among 14 counters. Multiply:  $\frac{7}{15} \times \frac{8}{14} = \frac{4}{15}$ .

# Practice worked solutions

## 5. $\frac{4}{15}$

P(red first) =  $\frac{8}{16}$ . Then 8 blue remain among 15 counters. Multiply:  $\frac{8}{16} \times \frac{8}{15} = \frac{4}{15}$ .

## 6. $\frac{7}{30}$

P(red first) =  $\frac{7}{10}$ . Then 3 blue remain among 9 counters. Multiply:  $\frac{7}{10} \times \frac{3}{9} = \frac{7}{30}$ .

## 7. $\frac{14}{55}$

P(red first) =  $\frac{7}{11}$ . Then 4 blue remain among 10 counters. Multiply:  $\frac{7}{11} \times \frac{4}{10} = \frac{14}{55}$ .

## 8. $\frac{45}{182}$

P(red first) =  $\frac{9}{14}$ . Then 5 blue remain among 13 counters. Multiply:  $\frac{9}{14} \times \frac{5}{13} = \frac{45}{182}$ .

# Practice worked solutions

## 9. $\frac{3}{10}$

P(red first) =  $\frac{3}{5}$ . Then 2 blue remain among 4 counters. Multiply:  $\frac{3}{5} \times \frac{2}{4} = \frac{3}{10}$ .

## 10. $\frac{1}{4}$

P(red first) =  $\frac{3}{9}$ . Then 6 blue remain among 8 counters. Multiply:  $\frac{3}{9} \times \frac{6}{8} = \frac{1}{4}$ .

## 11. $\frac{4}{15}$

P(red first) =  $\frac{8}{16}$ . Then 8 blue remain among 15 counters. Multiply:  $\frac{8}{16} \times \frac{8}{15} = \frac{4}{15}$ .

## 12. $\frac{35}{132}$

P(red first) =  $\frac{5}{12}$ . Then 7 blue remain among 11 counters. Multiply:  $\frac{5}{12} \times \frac{7}{11} = \frac{35}{132}$ .

# Practice worked solutions

## 13. $\frac{8}{33}$

P(red first) =  $\frac{4}{12}$ . Then 8 blue remain among 11 counters. Multiply:  $\frac{4}{12} \times \frac{8}{11} = \frac{8}{33}$ .

## 14. $\frac{4}{15}$

P(red first) =  $\frac{6}{10}$ . Then 4 blue remain among 9 counters. Multiply:  $\frac{6}{10} \times \frac{4}{9} = \frac{4}{15}$ .

## 15. $\frac{4}{15}$

P(red first) =  $\frac{2}{6}$ . Then 4 blue remain among 5 counters. Multiply:  $\frac{2}{6} \times \frac{4}{5} = \frac{4}{15}$ .

## 16. $\frac{35}{132}$

P(red first) =  $\frac{5}{12}$ . Then 7 blue remain among 11 counters. Multiply:  $\frac{5}{12} \times \frac{7}{11} = \frac{35}{132}$ .

# Transfer worked solutions

## 1. $\frac{7}{30}$

P(red first) =  $\frac{7}{10}$ . Then 3 blue remain among 9 counters. Multiply:  $\frac{7}{10} \times \frac{3}{9} = \frac{7}{30}$ .

## 2. $\frac{35}{132}$

P(red first) =  $\frac{7}{12}$ . Then 5 blue remain among 11 counters. Multiply:  $\frac{7}{12} \times \frac{5}{11} = \frac{35}{132}$ .

## 3. $\frac{5}{18}$

P(red first) =  $\frac{5}{10}$ . Then 5 blue remain among 9 counters. Multiply:  $\frac{5}{10} \times \frac{5}{9} = \frac{5}{18}$ .

## 4. $\frac{3}{10}$

P(red first) =  $\frac{2}{5}$ . Then 3 blue remain among 4 counters. Multiply:  $\frac{2}{5} \times \frac{3}{4} = \frac{3}{10}$ .

# Transfer worked solutions

## 5. $\frac{3}{10}$

P(red first) =  $\frac{3}{6}$ . Then 3 blue remain among 5 counters. Multiply:  $\frac{3}{6} \times \frac{3}{5} = \frac{3}{10}$ .

## 6. $\frac{9}{35}$

P(red first) =  $\frac{9}{15}$ . Then 6 blue remain among 14 counters. Multiply:  $\frac{9}{15} \times \frac{6}{14} = \frac{9}{35}$ .



# Exit-ticket worked solutions

## 1. $\frac{7}{26}$

P(red first) =  $\frac{7}{13}$ . Then 6 blue remain among 12 counters. Multiply:  $\frac{7}{13} \times \frac{6}{12} = \frac{7}{26}$ .

## 2. $\frac{3}{14}$

P(red first) =  $\frac{6}{8}$ . Then 2 blue remain among 7 counters. Multiply:  $\frac{6}{8} \times \frac{2}{7} = \frac{3}{14}$ .

## 3. $\frac{9}{44}$

P(red first) =  $\frac{9}{12}$ . Then 3 blue remain among 11 counters. Multiply:  $\frac{9}{12} \times \frac{3}{11} = \frac{9}{44}$ .

## 4. $\frac{35}{132}$

P(red first) =  $\frac{7}{12}$ . Then 5 blue remain among 11 counters. Multiply:  $\frac{7}{12} \times \frac{5}{11} = \frac{35}{132}$ .



# Use errors to choose the next step

## Identify the first mismatch

Compare the student's representation with the quantities or stated condition. Separate reading the situation, choosing a method, calculation, and interpretation.

## Respond precisely

If the model is wrong, use a simpler example before repeating the procedure. If the method is sound and arithmetic is wrong, correct that step and verify the original result.

## Review the exit ticket

For each item: 2 points for an accurate result with coherent reasoning; 1 for a sound method with a minor error or an unexplained correct answer; 0 for unsupported or missing reasoning. Use comments to select a next action. These are local instructional criteria.

## AIG

Compare two methods, construct a counterexample, or create a new problem with a prescribed solution and justify its restrictions.

## 504 / EC

Follow the individual plan. Offer manageable item groups, documented time or format supports, a worked step, or an alternate response method while retaining the mathematical objective.

## ESL

Preview terms and symbols in the questions. Use home-language rehearsal or a bilingual glossary, then ask for a concise explanation such as "The condition means \_\_\_, so I used \_\_\_."

# Sources, preparation, and license

Objective: Compute probabilities from equally likely outcomes. State alignment: not claimed. This is a focused college-entrance preparation resource, not a state course or a claim of course equivalency.

Official exam scope consulted September 15, 2026: College Board, Types of Math Tested, <https://satsuite.collegeboard.org/sat/whats-on-the-test/math/types> ; ACT, Mathematics Test Description, <https://www.act.org/content/act/en/products-and-services/the-act/test-preparation/description-of-math-test.html> .

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